TEXTILE TECHNOLOGY

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW



EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. FOURTH YEAR

TEXTILE TECHNOLOGY

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2021-22]

B.Tech. VII Semester
Textile Technology

S. No.	Course Code	Course Title	Periods			Evaluation Scheme				End Semester		Total	Credits
			L	T	P	CT	TA	Total	PS	TE	PE		
1.	KHU701 /KHU702	HSMC -1 #/HSMC-2 #	3	0	0	30	20	50		100		150	3
2.	KTT 071 KTT-072/	Garment Manufacture Technology/ Theory of Textile Structure/	3	0	0	30	20	50		100		150	3
3.	KTT074/ KTT075/ KTT076	Process Control in Spinning & Weaving/ Mill Planning & Organization/ Evaluation of Textile Materials	3	0	0	30	20	50		100		150	3
4.		Open Elective - II	3	0	0	30	20	50		100		150	3
5.	KTT-751	Garment Manufacture Technology	0	0	2				25		25	50	1
6.	KTT-752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7.	KTT-753	Project I	0	0	8				150			150	4
		MOOCs (Essential for Hons. Degree)											
		Total										**850**	**18**

B.Tech. VIII Semester
Textile Technology

S. No.	Course Code	Course Title	Periods			Evaluation Scheme				End Semeste		Total	Credits
			L	T	P	CT	TA	Total	PS	TE	PE		
1.	KHU701/K HU702	HSMC -1 #/HSMC-2 #	3	0	0	30	20	50		100		150	3
2.		Open Elective –III	3	0	0	30	20	50		100		150	3
3.		Open Elective –IV	3	0	0	30	20	50		100		150	3
4.	KTT851	Project II	0	0	18				100		300	400	9
		MOOCs (Essential for Hons.											
		Total										**850**	**18**

B.Tech 4rd Year
VII Semester
Syllabus

KTT-071	Garment Manufacturing Technology	3L:0T:2P	3 Credits

Course Objectives:

1. To familiarize with Indian garment industry and various processes involved in garment manufacturing.
2. To develop understanding of various types of commercial fabrics, trims and accessories used in garment manufacturing and their inspection methods.
3. To understand the concept of marker planning, fabric spreading and cutting and other pre-sewing processes.
4. To develop understanding of various types of stitches, seams, sewing defects, sewing threads and needles used in garment manufacturing.
5. To get familiarized with commonly used industrial sewing and finishing machines, apparel production systems and industrial engineering concepts.

Unit	Topics	Lectures
I	Introduction to Indian garment industry, overview of garment manufacturing processes, understanding of body measurements-Men's and Ladies Wear, familiarization with pattern making terminology.	8
II	Understanding of commercial woven and knitted fabrics, defects in fabrics due to weaving, knitting, and chemical processing, fabric inspection systems and feedback process; types of trims and accessories used in garment manufacturing.	8
III	Introduction to marker planning, efficiency of marker planning, methods of marker planning and marker use, fabric spreading and its requirements, methods of spreading, fabric packages, objective of cuttings, methods of cutting, fusing, **welding & moulding techniques**, ticketing and bundling processes.	8
IV	**I** Introduction to stitches and seams, methods of stitch formation, stitch and seam classification, stitch and seam defects, types of sewing needles and sewing thread, calculation of sewing thread consumption..	8
V	Classification of industrial sewing machines, knowledge of commonly used sewing machines in the garment industry, their major parts and functions, **Seam sealing machine for PPE Kits**, pressing and finishing machinery, assembly-line production system; time and method study, line balancing concept, Acceptable quality level.	8

Text Book:

1. Pattern Making for fashion design by Helen J. Armstrong
2. Commercial fabrics by Akshay Tholia
3. Technology of Clothing Manufacture By Barbara Carr and Harold Latham
4. Garment Manufacturing: Processes, Practices and Technology by Prasanta Sarkar
5. Industrial Engineering in Apparel Production by V. Ramesh Babu

Course Outcome: At the end of the course, students will develop adequate knowledge and skills to be able to work in garment industry as production executives, quality controllers or IE executives. They will be able to:

- Classify and Explain different fabrics and trims used in garment manufacturing
- Use spreading, cutting, sewing and finishing techniques in garment industry.
- Identify and explain various types of machinery used in garment manufacturing.
- Develop knowledge of different production systems and industrial engineering concepts used in garment industry.

KTT-072	Theory of Textile Structure	3L:0T:0P	3 Credits

Course Objectives:

1. To understand various aspects related to yarn geometry.
2. To understand the fibre migration, mechanism of migration and its impact on yarn strength.
3. To understand mechanism of staple fibre yarns, translation of fibre properties into yarn properties.
4. To understand the elements of fabric geometry, relation between cover and weight per unit area of fabric.

Unit	Topics	Lectures
I	Classification of yarns, Yarn geometry- idealized yarn geometry, relationship of yarn number and twist factor, packing of fibres in a yarn, ideal packing, hexagonal close packing and radial packing, packing factor and its measurement, yarn diameter, method of measurement of twist contraction.	8
II	Fibre migration: mean fibre position, amplitude of migration and frequency of migration, mechanism of migration, spinning-in coefficient and fibre extention, estimation of strength of yarns.	8
III	Mechanism of staple fibre yarns, translation of fibre properties into yarn properties, twist and strength relationship, limit of twist, spinability of textile fibres, relation with end-breakage rate.	8
IV	Elements of fabric geometry, cloth setting theories, flexible and rigid thread model, Pierce's equation and later modifications.	8
V	Relation of fabric properties to simple geometry, crimp interchange in woven fabric, crimp balance equation, Fabric cover, cover factor and their significance, relation between cover and weight per unit area of fabric, Theoretical treatment of fabric deformation in tension.	8

Text Book:

1. Textile Yarn- B.C. Goswami, J.G. Martindale, F.L. Scardine
2. Textile structure- J.W.S. Hearle, S. Backer, Grossberg.
3. Pierce's geometry- Textile institute

Course Outcome: At the end of the course, students will be able to:

1. Describe the fibre migration, mechanism of migration and its impact on yarn strength.
2. Explain mechanism of staple fibre yarns and correlate fibre properties and yarn properties.
3. Explain elements of fabric geometry, relation between cover factor, GSM etc.

KTT074	Process Control in Spinning & Weaving	3L:0T:0P	3 Credits

Course Objectives:

1. To understand importance of process control in various area of spinning process.
2. To understand the importance of productivity, means to improve productivity, maximizing machine efficiency in ring spinning,.
3. To understand scope and systems of process control in weaving, setting norms and schedule of checks, machinery audit.
4. To understand the process control in warping, minimizing end breakage in warping, quality of warping beam,
5. To understand the process control in loom shed, snap study and time and motion study, control of warp and weft breakage, causes and remedies of fabric defects.

Unit	Topics	Lectures
I	Importance of evolving a system for process control, control of mixing quality through fibre characteristics, simultaneous control of mixing cost and quality, concept of bale management. Control of waste in blow room and carding, norms for waste and cleaning efficiency in blow room & card, control of neps, assessment of performance of blow room & card, control of comber waste, concept of yarn realization, calculation pertaining to waste & yarn realization.	8
II	Measurement and analysis of productivity, means to improve productivity, maximizing machine efficiency in ring spinning, controlling end breakage rate in ring spinning, control of soft waste and hard waste, control of yarn faults and package defects- slubs, crackers, spinner's doubles, bad piecing & slough off. Control of yarn quality- count, strength and their variability, study of CV% (within bobbin and between bobbin), control of variability of lea strength, single yarn strength and elongation %, Control of yarn unevenness, imperfections and hairiness.	8
III	Scope of process control in weaving, systems of process control in weaving, setting norms and schedule of checks, machinery audit, optimizing quality of winding, control of quality of knot, producing good package, dressing of the beams for reducing incidence of cross ends.	8
IV	Process control in warping, minimizing end breakage in warping, quality of warping beam, control of productivity, factors responsible for loss in efficiency, control of productivity, Process control in sizing, choice recipe and size pick-up, control of size pick-up, control of yarn stretch, quality of sized beams.	8
V	Process control in loom shed, snap study and time and motion study, control of warp and weft breakage, causes and remedies of fabric defects, factors responsible for loss inefficiency, control of productivity, process control in grey inspection, and folding sections.	8

Text Book:

1. Machine catalogues of various machines manufacture
2. Process Control in Spinning by ATIRA
3. Process control in weaving by ATIRA
4. Process Control in Spinning by K.R. Salhotra

Course Outcome: At the end of the course, students will develop adequate knowledge and skills to be able to work in garment industry as production executives, quality controllers or IE executives. They will

be able to:

1. Explain the importance of process control in various area of spinning process.
2. Describe productivity, means to improve productivity, maximizing machine efficiency in ring spinning.
3. Explain process control in weaving, setting norms and schedule of checks, machinery audit.
4. Explain process control in warping, minimizing end breakage in warping, quality of warping beam,
5. Explain process control in loom shed, snap study and time and motion study, control of warp and weft breakage, causes and remedies of fabric defects.

KTT075	Mill Planning & Organization	3L:0T:0P	3 Credits

Course Objectives:

1. To understand various aspects related to preparation of project in spinning/weaving process.
2. To understand Industrial hazards, safety aspects, humidification and air conditioning, etc.
3. To understand the plant layout for machines- balancing of machines, etc.
4. To understand the various terms used in costing (cost volume, profit analysis, cost allocation on waste etc).
5. To understand the Economic Viability, Staff organization and MIS in textile mills.

Unit	Topics	Lectures
I	Preparation of project—Spinning/Weaving/Processing/Composite I- Selection of product J- Site Selection (site, location, land and cost) K- Building (single, double and multiple) L- Plant and Machinery	8
II	Industrial hazards: i. Fire hazards, ii. Mechanical hazards, iii. Electrical hazards etc Safety rules for prevention of accidents Humidification of textile mill- humidifier and humidification Ventilation, floor cleaning in textile mills, lightening Air conditioning and Refrigeration system	8
III	Balancing of machine (plant layout for machines-- balancing of machines, layout of different machines, calculation for balancing of machines for different processes—spinning, weaving)	8
IV	Production Costing—various terms used in costing (cost volume, profit analysis, cost allocation on waste, effect on cost direct, indirect. Various elements of costing- concept of estimation for costing, break-even analysis.	8
V	Economic Viability Staff organization in textile mills – daily wages, various systems Recruitment, allocation and skill development Management and information system MIS	8

Text Book:

1. Industrial Engineering, Organization & management by Tarachand
2. Industrial Economics & Principle of Management by T.M. Chabra
3. Industrial Economics & Principle of Management by S. K Sharma

Course Outcome: At the end of the course, students will be able to:

1. Explain various aspects related to preparation of project in spinning/weaving process.
2. Describe Industrial hazards, safety aspects, humidification and air conditioning, etc.
3. Explain the plant layout for machines- balancing of machines, etc.
4. Explain various terms used in costing (cost volume, profit analysis, cost allocation on waste etc).
5. Evaluate the Economic Viability, Staff organization and MIS in textile mills.

KTT076	Evaluation of Textile Materials	3L:0T:0P	3 Credits

Course Objectives:

1. To understand Sampling Methods and evaluation of Textile Materials.
2. To understand Evaluation of Fibre Fineness, Fibre Maturity, and Yarn Twist.
3. To understand the methods of evaluation of cotton fibre Properties, linear density moisture and yarn & fabric hairiness.
4. To understand the method of evaluation of tensile properties of textile materials
5. To understand the method of evaluation of yarn evenness, bursting & tear strength of fabrics, evaluation of pilling & abrasion properties of fabrics.

Unit	Topics	Lectures
I	Evaluation of Textile Materials, Sampling Methods & Sample Size, Sampling Methods & Sample Size: Practical Statistics, Evaluation of Fibre Length.	8
II	Evaluation of Fibre Fineness, Evaluation of Cotton Fibre Maturity, Evaluation of Yarn Twist.	7
III	Evaluation of Cotton Fibre Properties:(HVI & AFIS), Evaluation of Linear Density of Textile Materials, Evaluation of Moisture in Textiles, Evaluation of Yarn & Fabric Hairiness.	8
IV	Evaluation of Tensile Properties of Textile Materials, Evaluation of Tensile Properties of Textile Materials-1 , Evaluation of Tensile Properties of Textile Materials-2 , Evaluation of Tensile Properties of Textile Materials-3,4,5	9
V	Evaluation of Yarn Evenness-1-5 Evaluation of Bursting & Tear Strength of Fabrics, Evaluation of Pilling & Abrasion Properties of Fabrics, Evaluation of Low Stress Mechanical Properties of Textile Materials.	8

Text Book:

1. Industrial Engineering, Organization & management by Tarachand
2. Industrial Economics & Principle of Management by T.M. Chabra
3. Industrial Economics & Principle of Management by S. K Sharma

Course Outcome: At the end of the course, students will be able to:

1. Describe the sampling methods and evaluation of textile materials.
2. Explain evaluation of fibre fineness, fibre maturity, and yarn twist.
3. Demonstrate the methods of evaluation of cotton fibre Properties, linear density moisture and yarn & fabric hairiness.
4. Demonstrate the method of evaluation of tensile properties of textile materials
5. Demonstrate the method of evaluation of yarn evenness, bursting & tear strength of fabrics, evaluation of pilling & abrasion properties of fabrics.

KTT 751	Garment Manufacture Technology Lab	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

To understand body measurements of Men's and Ladies Wear; Develop understanding of different types of industrial sewing machines – Single needle Lockstitch Machine, Double needle Lockstitch Machine, Overlock, Flatlock and specialized computerised machines;- their machine parts and functions; Development of samples of different stitch types; Development of samples of different seam types; Understanding of different types of sewing needles; Development of samples of different types of sewing defects

References:

1. Stitches, TaBlets Series NITRA Publications
2. Seams, TaBlets Series NITRA Publications
3. Sewing Needles, Sewing Machine Series, NITRA Publications
4. Quality Assurance in Apparel Manufacturing, TaBlets Series NITRA Publications